



Title: Pancreatic Tumor Segmentation using Self-Supervised and Transfer Learning: PANTHER Challenge

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Problem Statement

Clinical & Technical Motivation:

Pancreatic cancer is one of the deadliest cancers due to late diagnosis and complex tumor morphology.

Accurate tumor segmentation in MRI is critical for:

Diagnosis & Staging (Task 1)

Adaptive Radiotherapy (Task 2)

Core Challenges:

MRI Complexity:

T1-weighted scans have variable contrast (arterial phase, venous phase, etc.).

T2-weighted MR-Linac scans are used in real-time, but annotated data is limited.

Data Scarcity:

Task 2 has only 50 annotated MRIs (few-shot learning needed).

Task 1 has 367 unlabeled scans—opportunity for self-supervised pretraining.

Task Overview – What Are the Two Tasks?

Task	MRI Type	Purpose	Main Challenge
Task 1	T1-weighted contrast-enhanced	Diagnosis & Staging	Variable contrast, subtle tumour visibility
Task 2	T2-weighted MR-Linac	Adaptive Radiotherapy	Few labeled samples, real-time constraints

MRI Types Explained

T1-weighted Arterial Contrast-Enhanced MRI (Task 1):

Captures **vascular structure** using contrast agents.

Tumour appears **hypo- or iso-intense**.

High variability in enhancement → difficult segmentation.

T2-weighted MRI (Task 2):

Emphasizes **fluid & soft tissue**.

Tumour typically **hyperintense**.

Ideal for treatment monitoring, but data limited.

Dataset Comparison

Feature	Task 1	Task 2
Annotated Scans	92	50
Unannotated Scans	367	0
MRI Type	Multi-phase T1	T2 MR-Linac
Use Case	Diagnosis	Radiotherapy
Challenge	Contrast variation	Few-shot segmentation

Technical Workflow – Overview

1. Preprocessing (Resampling, Normalization)
2. Self-Supervised Pretraining (SimCLR / MAE on T1)
3. Supervised Fine-Tuning (on annotated T1 data)
4. Transfer Learning to Task 2
5. Few-shot learning + Data Augmentation
6. Evaluation & Post-processing

Step-by-Step – Task 1 Pipeline

Preprocessing:

Normalize intensity (z-score), resize, align orientation

◆ Self-Supervised Pretraining:

Use **MAE** or **SimCLR** on 367 unannotated multi-phase T1 MRIs

Output: encoder pretrained on pancreas anatomy

◆ Fine-Tuning:

Architecture: **Swin UNETR** / **EfficientUNet**

Loss: Dice + BCE

Use 92 annotated images for training

Step-by-Step – Task 2 Pipeline

- ◆ **Transfer Learning:**

Load encoder from Task 1

- ◆ **Few-Shot Setup:**

Train on just 50 T2 MRIs

Model: TransUNet or UNETR with attention

- ◆ **Augmentation:**

Elastic deformation, affine, noise, dropout

- ◆ **Optional Semi-Supervision:**

Pseudo-labeling + consistency regularization (if more data available)

Model Architecture Choices

Component	Model
SSL Pretraining	MAE / SimCLR / BYOL
Task 1 Segmentation	Swin UNETR / EfficientUNet / nnUNet
Task 2 Adaptation	TransUNet / R50-ViT / Lightweight UNETR
Data Tools	MONAI, TorchIO, NiBabel

Evaluation Metrics

1. Dice Score – Overlap of prediction & ground truth
2. IoU (Jaccard Index) – Intersection over union
3. Hausdorff Distance – Boundary-based precision
4. Cross-validation – 5-fold for robustness

Accuracy Boosting Techniques

1. Multi-view Training (Axial, Coronal, Sagittal)
2. Curriculum Learning (Train easy → hard)
3. CRF or Morphological Post-processing
4. Model Ensembling (blend 2–3 models)
5. Co-Training (joint pancreas + tumour models)

Project Summary & Future Scope

Achievements:

Designed robust segmentation pipeline using SSL and transfer learning

Addressed data scarcity with advanced augmentation

Improved segmentation in real-time MRI therapy context

Future Work:

Domain generalization across hospitals

Integrate temporal sequences (pre-/post-treatment)

Real-time segmentation on MR-Linac during therapy

References

PANTHER Challenge Dataset ([Link](#))

SimCLR Paper ([Link](#))

MAE Paper ([Link](#))

Swin UNETR ([Link](#))

MONAI Documentation ([Link](#))